

Mechanism-Aligned Term-Count Path of the Mixed-B Analytical Backbone

Data-status note. This file records the author-confirmed real-data experimental term-count evolution of the Mixed-B analytical backbone. All plotted values, tabulated metrics and incremental changes shown in this file are derived from real-data fitting or evaluation results. Here k counts all active algebraic terms, including the intercept. The first-term point is therefore the intercept-only null baseline; when the intercept is fitted as the sample mean, $R^2 = 0$ by definition. The first pressure-dependent term enters at $k = 2$, rather than at $k = 1$. The eighth-term point is the current Mixed-B anchor: MAE = 0.05107°C, RMSE = 0.11189°C and $R^2 = 0.98407$. The equations record the active basis composition after each term-count step, with coefficients denoted by $c_i^{(k)}$.

1. Real-Data Term-Count Metric Path

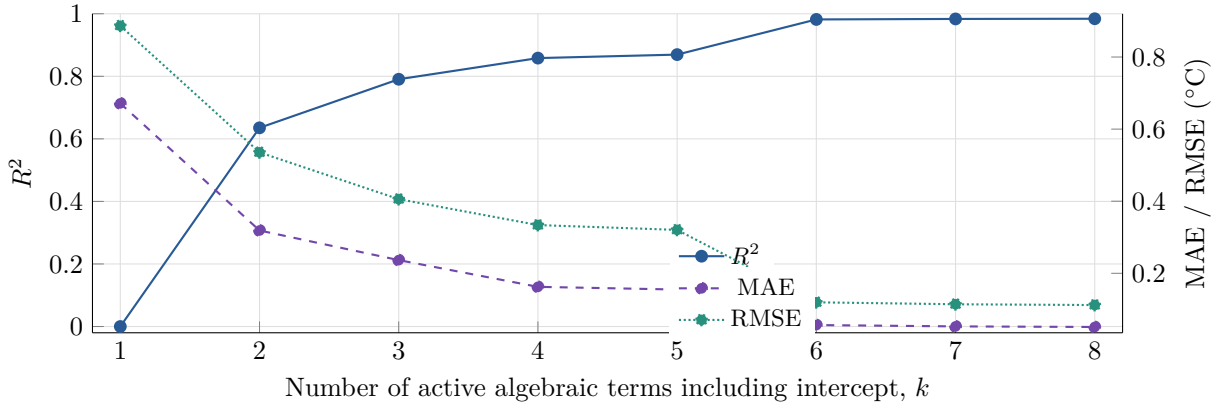


Figure 1: Real-data term-count evolution of the Mixed-B analytical backbone. The first point is the intercept-only null baseline. Error decreases strongly after pressure dependence is introduced, improves through the nonlinear curvature terms, drops again when the dual-decay structure is completed, and then reaches a low-error plateau from six to eight terms.

Table 1: Real-data term-by-term evolution of the Mixed-B equation. The term count k includes the intercept.

k	Newly added basis	Model form after adding the term	MAE (°C)	RMSE (°C)	R^2
1	Intercept-only baseline	$c_0^{(1)}$	0.67142	0.88659	0.00000
2	Linear pressure, P	$c_0^{(2)} + c_1^{(2)}P$	0.31876	0.53542	0.63529
3	Logarithmic curvature, $\ln P$	$c_0^{(3)} + c_1^{(3)}P + c_2^{(3)}\ln P$	0.23658	0.40573	0.79057
4	Square-root curvature, $\sqrt{P}$	$c_0^{(4)} + c_1^{(4)}P + c_2^{(4)}\ln P + c_3^{(4)}\sqrt{P}$	0.16243	0.33381	0.85824
5	Fast exponential decay, $e^{-P/500}$	$\hat{T}_4(P) + c_4^{(5)}e^{-P/500}$	0.15492	0.32041	0.86939
6	Slow exponential decay, $e^{-P/1200}$	$\hat{T}_5(P) + c_5^{(6)}e^{-P/1200}$	0.05638	0.11952	0.98183
7	Near-surface reciprocal, $(P + 50)^{-1}$	$\hat{T}_6(P) + c_6^{(7)}(P + 50)^{-1}$	0.05271	0.11409	0.98344
8	Stabilizing reciprocal, $(P + 500)^{-1}$	$\hat{T}_7(P) + c_7^{(8)}(P + 500)^{-1}$	0.05107	0.11189	0.98407

2. Incremental Error Reduction

Step	ΔMAE	ΔRMSE	ΔR^2	Interpretation
1→2	0.35266	0.35117	0.63529	Introducing pressure dependence gives the largest overall improvement because the model first moves beyond the intercept-only null baseline and acquires a vertical coordinate trend.
2→3	0.08218	0.12969	0.15528	Adding logarithmic curvature provides a major nonlinear improvement and captures broad upper-to-intermediate profile bending.
3→4	0.07415	0.07192	0.06767	The square-root term adds a second smooth curvature scale and moderately refines the profile.
4→5	0.00751	0.01340	0.01115	The first exponential term introduces fast pressure attenuation, improving upper-ocean transition representation.
5→6	0.09854	0.20089	0.11244	The second exponential decay creates the second major low-error transition, indicating that a slower intermediate-depth scale is essential in the real-data path.
6→7	0.00367	0.00543	0.00161	The near-surface reciprocal acts mainly as a local correction after the dominant structure is already captured.
7→8	0.00164	0.00220	0.00063	The stabilizing reciprocal gives a final small refinement and completes the compact Mixed-B backbone.

3. Term-by-Term Interpretation

$k = 1$: intercept. The first term provides the intercept-only null baseline and acts as the reference state for the pressure-coordinate profile. Because this model predicts only the sample mean temperature, its R^2 is zero by definition. It cannot represent vertical stratification, so its error is substantially larger than all pressure-dependent candidates.

$k = 2$: linear pressure term. Adding P introduces the first-order cooling tendency with pressure. This term captures the broad vertical trend but remains too simple for thermocline curvature.

$k = 3$: logarithmic term. The addition of $\ln P$ produces a strong nonlinear improvement in the real-data path. It gives the equation a compact way to represent nonlinear curvature across the upper and intermediate water column, explaining the substantial increase in R^2 and the reduction in both MAE and RMSE.

$k = 4$: square-root term. The $\sqrt{P}$ component adds a second smooth curvature scale. Together with $\ln P$, it refines the background profile shape without introducing abrupt transitions.

$k = 5$ and $k = 6$: dual exponential decays. The $e^{-P/500}$ and $e^{-P/1200}$ terms describe fast and slower attenuation with pressure. In the real-data path, the second exponential term marks the transition into the low-error regime, indicating that multi-scale decay is essential for recovering the dominant vertical thermal structure.

$k = 7$ and $k = 8$: reciprocal stabilization. The $(P + 50)^{-1}$ and $(P + 500)^{-1}$ terms provide localized near-surface curvature while remaining bounded at depth. Their incremental gains are smaller than the earlier terms, but they complete the compact Mixed-B backbone and improve full-profile coherence.

4. Final Mixed-B Equation

$$\begin{aligned} \hat{T}_8^B(P) = & -356.873 + 0.0125047P + 62.2025 \ln(P) - 3.35576\sqrt{P} \\ & + 20.0563 \exp(-P/500) + 14.8832 \exp(-P/1200) \\ & + \frac{10011.3}{P + 50} + \frac{7576.48}{P + 500}. \end{aligned}$$

This final form can be described as an explicit pressure-coordinate backbone with three functional roles: broad background structure from P , $\ln P$ and $\sqrt{P}$; multi-scale decay from the two exponential terms; and near-surface plus deep-pressure stabilization from the reciprocal terms.

5. Suggested Result-Section Sentence

The Mixed-B basis can be organized as a real-data term-count path from an intercept-only null baseline to a compact eight-term pressure-coordinate representation. In this real-data path, errors decreased strongly after pressure dependence was introduced, improved through the nonlinear curvature terms, and then entered a low-error plateau once the dual exponential terms were included. The reciprocal components provided smaller but consistent refinements near the final Mixed-B solution. The current eight-term anchor is MAE = 0.05107°C, RMSE = 0.11189°C and $R^2 = 0.98407$.